

RANSHOFEN, Austria

WMO: 110510

Lat: 48.224N

Lon: 13.028E

Elev: 388

StdP: 96.75

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-8.2	-13.7	1.2	-10.4	-10.7	1.6	-7.3	7.2	3.4	6.4	3.1	1.9	100	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	30.8	20.5	28.8	19.8	27.0	19.0	21.2	28.8	20.4	27.2	19.6	25.6	2.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.3	23.8	18.1	13.6	22.9	17.4	13.0	22.3	63.4	29.1	60.3	27.4	57.6	25.7	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.4	4.6	-13.4	34.1	3.7	1.2	-16.0	35.0	-18.2	35.7	-20.3	36.4	-23.0	37.3		
(5)			-14.0	23.0	3.6	1.0	-16.7	23.8	-18.8	24.4	-20.8	25.0	-23.5	25.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-0.5	1.2	5.3	10.4	14.5	18.0	19.5	19.3	14.7	10.0	4.4	1.0	(6)	
	DBStd	8.04	4.26	4.36	4.01	4.05	3.75	3.78	3.34	3.34	3.28	3.54	3.67	3.67	(7)	
	HDD10.0	1273	327	247	149	44	6	1	0	0	3	46	172	278	(8)	
	HDD18.3	3276	585	480	403	240	128	52	27	29	117	259	419	537	(9)	
	CDD10.0	1220	0	1	5	55	146	240	296	287	143	44	3	0	(10)	
	CDD18.3	181	0	0	0	1	9	41	64	58	7	0	0	0	(11)	
	CDH23.3	1611	0	0	0	18	109	378	553	496	55	2	0	0	(12)	
	CDH26.7	460	0	0	0	2	17	100	175	161	6	0	0	0	(13)	
(14) Wind	WSAvg	2.0	2.2	2.3	2.4	2.2	2.2	1.9	1.8	1.7	1.8	2.0	2.0	2.1	(14)	
(15) Precipitation	PrecAvg	1036	61	54	77	61	106	129	127	120	96	71	69	67	(15)	
	PrecMax	1265	136	157	145	174	191	256	265	262	192	135	157	118	(16)	
	PrecMin	736	4	8	20	0	15	45	17	37	37	13	2	11	(17)	
	PrecStd	123	36	35	39	38	47	50	55	50	37	32	33	34	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.1	19.2	25.8	28.7	32.0	33.2	33.5	27.2	22.4	16.5	11.8	(19)	
		MCWB	8.6	8.9	11.3	15.3	18.9	21.3	21.5	20.6	19.1	15.5	11.8	8.9	(20)	
	2%	DB	9.3	11.1	16.4	22.6	26.3	29.5	30.7	30.7	24.9	19.2	13.2	9.4	(21)	
		MCWB	6.8	7.3	10.2	14.0	17.7	20.2	20.6	20.3	18.1	14.4	10.1	7.4	(22)	
	5%	DB	7.1	9.1	14.0	20.4	24.2	27.4	28.5	28.2	22.7	17.2	11.1	7.6	(23)	
		MCWB	5.4	6.4	9.2	13.0	16.6	19.2	19.7	19.8	17.1	13.6	9.2	5.9	(24)	
	10%	DB	5.3	7.1	11.8	18.1	21.9	25.5	26.5	26.0	20.6	15.2	9.5	5.9	(25)	
		MCWB	3.9	4.9	8.0	11.8	15.1	18.2	19.0	18.7	16.0	12.8	7.9	4.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	9.4	12.1	16.1	19.9	22.3	22.5	22.1	19.9	16.5	12.2	9.3	(27)	
		MCDB	11.2	13.2	17.6	23.6	27.2	29.8	30.3	30.6	25.4	20.5	15.5	11.2	(28)	
	2%	WB	7.0	7.9	10.6	14.6	18.2	21.0	21.2	21.0	18.6	15.2	10.6	7.7	(29)	
		MCDB	8.8	10.8	15.3	21.5	25.1	27.9	29.0	28.5	23.8	18.2	12.8	9.3	(30)	
	5%	WB	5.6	6.5	9.5	13.4	17.0	19.9	20.3	20.1	17.6	14.1	9.4	6.0	(31)	
		MCDB	7.1	8.8	13.5	19.6	22.7	26.1	27.0	26.8	21.8	16.6	11.0	7.2	(32)	
	10%	WB	4.0	5.1	8.4	12.2	15.8	18.9	19.4	19.3	16.7	13.0	8.0	4.8	(33)	
		MCDB	5.3	7.0	11.4	17.5	20.9	24.2	25.0	25.1	20.0	15.1	9.4	6.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	6.6	8.6	10.2	9.8	9.9	10.0	10.1	9.0	7.6	5.1	4.6	(35)	
		MCDBR	7.6	10.4	13.5	14.7	14.2	13.8	13.9	14.2	12.4	11.1	8.3	7.6	(36)	
	5% WB	MCWBR	5.6	6.9	7.8	7.2	6.3	5.5	5.2	5.3	6.1	6.6	5.7	5.6	(37)	
		MCWBR	7.2	9.8	12.5	13.9	13.0	12.5	12.5	12.8	11.2	10.0	7.7	7.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.301	0.326	0.365	0.397	0.406	0.410	0.416	0.410	0.384	0.353	0.326	0.299	(40)	
		taud	2.409	2.346	2.303	2.250	2.266	2.292	2.272	2.307	2.349	2.422	2.453	2.444	(41)	
	Ebn at Noon		770	826	846	851	857	854	842	830	816	785	734	727	(42)	
		Edh at Noon	67	88	109	126	129	127	128	118	103	82	65	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.08	2.01	3.06	4.26	4.93	5.44	5.33	4.69	3.37	2.18	1.14	0.84	(44)	
	RadStd		0.17	0.27	0.37	0.54	0.59	0.50	0.51	0.51	0.45	0.24	0.15	0.11	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (86 neighbors)		+0.46	N/A	N/A	N/A	+0.34	+0.35	N/A	N/A	+61	+25	

Nomenclature: See separate page